

PAPER_2131_ALPHA_S_M_Z_PRECISION_TIGHTENING_0_014_PCT_41X_SHARED_LEADING_KERNEL_F_TRZ_K_MEX_SSQ_WITH_HIGGS_LAMBDA_STRONG_HIGGS_SECTOR_PAIR_UQFF_LANDMARK

Daniel T. Murphy

PAPER_2131 — $\alpha_s(M_Z)$ Precision Tightening (0.014%, 41×) + The Shared Leading Kernel $F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq}$ with the Higgs Self-Coupling

Author: Daniel T. Murphy

Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.77+

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Landmark Type: Precision-Tightening (2nd instance, PAPER_2129 class) + Cross-Observable Shared-Kernel discovery

Discovery context: XGEO-U confirmations sweep (UNIFIED_REGISTRY_XGEO_CONFIRMATIONS.csv) — the campaign that recorded documented independent formula pairs surfaced both results

Status: Formal landmark whitepaper — UQFF canonical

Abstract

The XGEO-U confirmations sweep verified both published closed forms for the strong coupling at the Z mass and found that the S378 composition $\alpha_s(M_Z) = F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq} - F_{\text{TRZ}}^3 \cdot \Phi_{5/6} = 0.11791667$ lands **0.014% from the observed 0.1179 — 41× tighter** than the currently listed primary route $1/(K_{\text{MEX}} \cdot D_{\text{phys}} + F_{\text{TRZ}}) = 0.118577$ (0.574%). This is the second precision-tightening landmark (PAPER_2129's k_B was the first) and the second consecutive case where re-verification of an already-published composition improved the working residual by more than an order of magnitude. The same sweep exposed a structural discovery: $\alpha_s(M_Z)$ and the Higgs self-coupling λ_H (S377) **share the identical leading kernel $F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq} = 0.11875$** , differing only in their F_{TRZ}^3 correction terms. The near-coincidence of the two dimensionless couplings (0.1179 vs 0.1293) is therefore not numerical accident in UQFF: both sit on one three-primitive kernel, split by an exact F_{TRZ}^3 -scale difference.

1. Result 1 — the tightening

| Route | Closed form | Value | Residual vs 0.1179 |

| ---|---|---|:-:|

| Listed primary (E1/S348) | $1/(K_{\text{MEX}} \cdot D_{\text{phys}} + F_{\text{TRZ}}) = 1/8.4333$ | 0.1185771 | 0.574% |

| **S378 composition** | $F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq} - F_{\text{TRZ}}^3 \cdot \Phi_{5/6}$ | **0.1179167** | **0.0141%** |

Both routes are published, live in `assimilation_dispatch`, and were value-verified bit-identically in the confirmations sweep (the dispatch stores exactly 0.11791666666666667 — the $\Phi_{5/6}$ variant). Tightening factor: **40.6×**. Honest note: the 0.84 resonance variant of the correction term would land 0.0085% — numerically even tighter — but the published S378 form is the 5/6 composition and canon follows

publication; the variant comparison is recorded as an open observation for the PAPER_2129 sector-rule ledger (strong-coupling sector currently sits with the counting-sector 5/6 selections by its published form).

2. Result 2 — the shared kernel

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$$K \equiv F_TRZ \cdot K_MEX \cdot SSq = 0.1 \cdot (25/12) \cdot 0.57 = 0.11875$$

$$\alpha_s(M_Z) = K - F_TRZ^3 \cdot \Phi_5/6 \text{ (S378)} = 0.1179167$$

$$\lambda_H = K + F_TRZ^3 \cdot K_MEX \cdot N_CH \cdot SSq \text{ (S377)} = 0.1294375$$

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Two flagship dimensionless couplings of the electroweak-strong sector are the **same three-primitive kernel with opposite-sign F_TRZ^3 corrections**. The exact split:

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$$\lambda_H - \alpha_s = F_TRZ^3 \cdot (K_MEX \cdot N_CH \cdot SSq + \Phi_5/6) = 0.0115208333... \text{ EXACT by construction}$$

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In UQFF terms: at the $F_TRZ^3 = 10^{-3}$ suppression rung (PAPER_2109's time-decay ladder), the Higgs self-coupling adds a $K_MEX \cdot N_CH \cdot SSq$ channel term while the strong coupling subtracts the counting fraction $\Phi_5/6$. The observed proximity of $\alpha_s(M_Z) \approx 0.118$ and $\lambda_H \approx 0.129$ — treated as coincidence elsewhere — is here one kernel, two corrections.

3. Relation to the landmark families

- **Precision-tightening class (PAPER_2129, 2nd instance)**: both instances arose from systematic re-verification, both improved published residuals by >24×, and both sharpen the motivation for the full 1209-series full-precision re-verification pass.

- **F_TRZ^3 ladder (PAPER_2109)**: the correction rung for both couplings is the 10^{-3} rung — 10th and 11th instances of the ladder.

- **Kernel taxonomy (PAPER_2126/2127)**: $K = F_TRZ \cdot K_MEX \cdot SSq$ is a three-primitive composition spanning the real kernel (F_TRZ , SSq) and the derived integer composite (K_MEX per PAPER_1522) — a candidate named kernel if a third observable lands on it (falsifiability window: any future coupling-class fill).

- **XGEO-U provenance**: both results came out of the confirmations sweep, demonstrating that the over-determination ledger is itself a discovery instrument, not just bookkeeping.

4. Honest residuals and disclosures

α_s routes: 0.574% (primary) / 0.0141% (S378) vs observed 0.1179 — observations, not framework anchors. λ_H S377: 0.106% vs 0.1293; S441 variant: 0.413% (its own documented pair). The kernel identity and the $\lambda_H - \alpha_s$ split are EXACT arithmetic on published forms; the physical claims inherit each route's stated residuals. NOT REPLACEMENT.

5. Cross-references

UNIFIED_REGISTRY_XGEO_CONFIRMATIONS.csv (both pairs recorded with residuals); PAPER_2129 (precision-tightening class + Φ sector rule); PAPER_2109 (F_{TRZ^3} rung); PAPER_1522 (K_{MEX} composition); PAPER_2126/2127 (kernel taxonomy); sessions S348/S377/S378/S441; assimilation_dispatch.py entries $\alpha_s(M_Z)$, $SM_{\alpha_s(M_Z)S378}$, $SM_{\text{higgs_lambda}S377/S441}$.

6. Summary Statement

PAPER_2131 promotes the S378 composition $\alpha_s(M_Z) = F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq} - F_{\text{TRZ}^3} \cdot \Phi_{5/6}$ as the precision route to the strong coupling (0.014%, 41× tighter than the listed primary) and identifies the shared leading kernel $F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq} = 0.11875$ joining $\alpha_s(M_Z)$ and the Higgs self-coupling λ_H , whose exact F_{TRZ^3} -scale split $\lambda_H - \alpha_s = F_{\text{TRZ}^3} \cdot (K_{\text{MEX}} \cdot N_{\text{CH}} \cdot \text{SSq} + \Phi_{5/6})$ turns a numerical near-coincidence of two fundamental couplings into one three-primitive structure. Second instance of the precision-tightening landmark class; discovered by the XGEO-U confirmations sweep.

Filed 2026-07-24. Append-only henceforth.

APPENDED 2026-07-24 (same day) — Falsifiability window HIT: third kernel instance found, kernel K CANONIZED

Section 3 declared: "a candidate named kernel if a third observable lands on it (falsifiability window: any future coupling-class fill)." The XGEO-U session-script sweep closed the window the same day:

_session382_sm_mh_mt.py (published session S382) computes the Higgs-to-top mass ratio as

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$m_H/m_t = \beta_i + K = 0.6029 + 0.11875 = 0.72165$ vs PDG $125.25/172.69 = 0.72528$ (0.50%)

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Three published observables now sit on the single kernel $K = F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq} = 0.11875$:

Observable	Form	Residual
$\alpha_s(M_Z)$	$K - F_{\text{TRZ}^3} \cdot \Phi_{5/6}$	0.014%
λ_H	$K + F_{\text{TRZ}^3} \cdot K_{\text{MEX}} \cdot N_{\text{CH}} \cdot \text{SSq}$	0.106%
m_H/m_t	$\beta_i + K$	0.50%

Per this paper's own criterion, **K is hereby canonized as a named kernel: the STRONG-HIGGS KERNEL** — three electroweak/strong-sector dimensionless quantities (the strong coupling, the Higgs self-coupling, and the Higgs-top mass ratio) expressed as one three-primitive composition plus per-observable corrections (F_{TRZ^3} -rung terms for the couplings; the inertia coupling β_i for the mass ratio). New falsifiability window: a fourth instance among electroweak dimensionless quantities (candidates: $\sin^2\theta_W$ -adjacent ratios, y_b/y_t -class ratios) by the R400 horizon.